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OCP MHS Modular Plug-and-Play Hardware Component Schema Documentation

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L = Local, **R** = Required

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Top-level Properties

Description: The HardwareComponent FRU schema defines the hardware properties of an HPM or of a component assembly such as a backplane or riser that are necessary to enable plug-and-play management of the component by a DC-SCM module attached to the HPM or to an HPM bearing the component.

Property	Type	L	R	Description
_comment	string			A general comment about a data file
Copyright	string			A copyright statement for the FRU data.
Version	string			The current file version for this FRU data.
SchemaVersion	string			The revision of the HardwareComponent schema that this FRU data was built against.
GitHubChecksum	string			The GitHub checksum for this FRU data.
Author	string			The author of this FRU data.
Component	Top-level Properties.Component			Information about a hardware component.

Top-level Properties.Component

Description: Information about a hardware component.

Property	Type	L	R	Description
ComponentClass	string	Y	Y	The type of hardware that this component represents. Possible values: `HPM` (A host processor module, such as a DNO, FLW, or SDNO board),

Property	Type	L	R	Description
				`Riser` (A riser), `Backplane` (A backplane), `PowerDistributionBoard` (A power distribution board), `PCIeSwitch`, `Other` (Any other component)
ComponentSubclass	string	Y	Y	The subtype of hardware that this component represents, such as DNO, FLW, or SDNO for an HPM.
VendorID	string	Y	Y	The vendor ID assigned by the manufacturer for this component.
UniqueID	string	Y	Y	A unique ID assigned by the manufacturer for this component, such as a PCIe DeviceID.
PowerPolicy	string			The minimum system power state in which the component can be powered for full operation. Possible values: `G3`, `Pre-STBY`, `STBY`, `S5`, `S0`
MaximumPowerWatts	integer			The maximum wattage that the component can draw when fully powered.
PhysicalDimensions	Top-level Properties.Component.PhysicalDimensions			Physical dimensions of the PCB for the component.
Connectors	Connectors			The hardware connectors that are present.
Devices	Devices			Information about active and passive hardware devices
Buses	Buses			Information about the logical and physical buses.

Top-level Properties.Component.PhysicalDimensions

Description: Physical dimensions of the PCB for the component.

Property	Type	L	R	Description
PCBWidthMM	integer			The width of the PCB in millimeters.
PCBLengthMM	integer			The length of the PCB in millimeters.

Connectors

Description: The hardware connectors that are present.

Property	Type	L	R	Description
SOCs	array of Connectors_SOCs			Information about SOC connectors.
MemorySubsystems	array of Connectors_MemorySubsystem			Information about memory subsystem connectors.
Composites	array of Connectors_Composite			Information about composite connectors.
CoolingSubsystems	array of Connectors_CoolingSubsystem			Information about cooling subsystem connectors.
Drives	array of Connectors_Drive			Information about drive connectors.
Mxios	array of Connectors_Mxio			Information about Mxio connectors.
PICPWRs	array of Connectors_PICPWR			Information about PICPWR connectors.

Property	Type	L	R	Description
OCPMezzanineSlots	array of Connectors_OCPMezzanineSlot			Information about OCP mezzanine expansion slot connectors.
PowerSupplies	array of Connectors_PowerSupply			Information about power supply connectors.
ControlPanels	array of Connectors_ControlPanel			Information about control panel connectors.
PCIe_CEMs	array of Connectors_PCIe_CEM			Information about PCIe CEM slot connectors.
PowerDistributionBoardManagements	array of Connectors_PowerDistributionBoardManagement			Information about power distribution board management connectors.
RealTimeClockBatteries	array of Connectors_RealTimeClockBattery			Information about real-time clock battery connectors.
Fans	array of Connectors_Fan			Information about fan connectors.
SCIs	array of Connectors_SCI	Y	Y	Information about SCI connectors.
IntrusionDetection	array of Connectors_IntrusionDetection			Information about intrusion detection module connectors.
PhysicalUSBs	array of Connectors_PhysicalUSB			Information about physical USB connectors.

Property	Type	L	R	Description
DB9s	array of Connectors_DB9			Information about DB9 connectors.
OCPBoots	array of Connectors_OCPBoot			Information about OCP boot connectors.
DisplayPorts	array of Connectors_DisplayPort			Information about DisplayPort connectors.
OEM	array of Connectors_OEM			Provides information for an OEM-specific connector.

Connectors_Composite

Description: Information about a composite connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Type	string			The type of the composite.
DataFile	integer			The data file with JSON information specific to this composite.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
PICPWRs	array of string	Y	Y	The PICPWRs used by the composite.
MXIOs	array of string	Y	Y	The MXIOs used by the composite.

Property	Type	L	R	Description
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_ControlPanel

Description: Information about a control panel connector.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Primary	boolean			Indicates whether this is the primary control panel.
Sidebands	SidebandList			A set of sidebands.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_CoolingSubsystem

Description: Information about a cooling subsystem connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
MaximumPowerWatts	integer			The maximum power for this cooling subsystem.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_DB9

Description: Information about a DB9 connector.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_DisplayPort

Description: Information about a DisplayPort connector.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.

Property	Type	L	R	Description
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_Drive

Description: Information about a drive connector.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
DriveType	string			The specific connector type for this drive, such as EDSFF or M.2.

Property	Type	L	R	Description
HotPlugSupported	boolean			Indicates whether this drive connector is wired to support hot plug.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_Fan

Description: Information about a fan connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Pins	integer			The number of pins for the connector for this fan.
MaximumPowerWatts	integer			The maximum power for this fan.
Type	string	Y	Y	The type of connector for this fan.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in

Property	Type	L	R	Description
				the BOM list.
HotPlugSupported	boolean			Indicates whether this fan supports hot plugging.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_IntrusionDetection

Description: Information about an intrusion detection module connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Type	string	Y	Y	The type of connector for this intrusion detection module.

Property	Type	L	R	Description
StatusSignalNetname	string			The netname of the signal that provides the state of the intrusion detection sensor.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.

Connectors_MemorySubsystem

Description: Information about a memory subsystem connector.

Property	Type	L	R	Description
ChannelCount	integer			The number of channels supported by the memory subsystem.
SlotsPerChannel	unknown			The number of slots in each channel supported by the memory subsystem.
MemoryTechnology	string			The memory technology supported by the memory subsystem. Possible values: `DDR4` (DDR4 SDRAM memory technology as defined in JEDEC specification JESD-4D.), `DDR5` (DDR5 SDRAM memory technology as defined in JEDEC specification JES-5C.01.), `DDR6` (DDR5 SDRAM memory technology as defined in a forthcoming JEDEC specification.),

Property	Type	L	R	Description
				`Other` (Any other memory technology.)
Slots	array of Connectors_MemorySubsystem.Slots	Y	Y	Information about an individual channels slot in the memory subsystem.

Connectors_MemorySubsystem.Slots

Description: Information about an individual channels slot in the memory subsystem.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Proximity	string			The proximity of the memory subsystem slot to its associated SOC. Possible values: `Near` (Close to the SOC.), `Far` (Further from the SOC, such that memory access performance is degraded.)
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
AssociatedSOC	string			The SOC associated with this memory subsystem slot.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Property	Type	L	R	Description
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.

Connectors_Mxio

Description: Information about an Mxio connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SOCSocketAssociation	string			The SOC Socket to which this MXIO is connected.
CommonClockingSupport	boolean			Indicates whether the MXIO supports common clocking.
Type	string	Y	Y	The type of connector for this MXIO.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
FlexIOs	array of FlexIO	Y	Y	The FlexIOs used by the MXIO.
Signals	array of Signal			Information about the signals for this MXIO.

Property	Type	L	R	Description
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_OCPBoot

Description: Information about an OCP boot connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
Type	string	Y	Y	The type of for the OCP boot connector.
Supports3_3V_AUX	boolean			Indicates whether the OCP boot connector supports 3.3V AUX power.
DualPortEnableSignalNetlistName	string			The netlist name for the dual port enable signal for the OCP boot connector.
LEDSignalNetlistName	string			The netlist name for the LED signal for the OCP boot connector.

Property	Type	L	R	Description
PowerDisableSignalNetlistName	string			The netlist name for the power disable signal for the OCB boot connector.
SMResetNetlistName	string			The netlist name for the SM bus reset signal for the OCP boot connector.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
PhysicalLocation	PhysicalLocation			The physical location of a component board.
FlexIOs	array of FlexIO			The FlexIOs used by the OCP Boot Connector.

Connectors_OCPMezzanineSlot

Description: Information about an OCP mezzanine expansion slot connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Version	string			The version of the OCP NIC specification with which this OCP mezzanine expansion slot complies.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
FormFactor	string			The form factor for this OCP mezzanine expansion slot. Possible

Property	Type	L	R	Description
				values: `SFF` (The Small Form Factor as defined in the OCP NIC 3.0 specification.), `TSFF` (The tall Small Form Factor as defined in the OCP NIC 3.0 specification.), `Other` (Any other form factor.)
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
PairedSlotForDoubleWide	string			The slot with which this slot is paired for use as a double-wide connector in the DSFF or TDSFF form factor.
Signals	array of Signal			Information about the signals for this OCP Mezzanine slot.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_OEM

Description: Information about an OEM connector.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.

Property	Type	L	R	Description
Type	string			The type of this OEM connector.
PresenceDetectSignal	string			Netlist name of the signal (e.g., a GPIO) used to detect presence of a device connected via this non-standard connector.
PresenceDetectPolarity	string			Polarity indicating the active/present state of PresenceDetectSignal. Possible values: `active_high`, `active_low`
MaximumPowerWatts	integer			The maximum power draw for a device connected via this connector.
RequiredPowerState	RequiredPowerState			The minimum power state to utilize a bus.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_PCIe_CEM

Description: Information about a PCIe CEM slot connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
Type	string	Y	Y	The type for this CEM slot.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
Signals	array of Signal			Information about the signals for this PCIe CEM slot.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_PICPWR

Description: Information about an PICPWR connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Type	string			The type of connector for this PICPWR.
AdjustedMaximumActualPowerSupportedWatts	integer			The adjusted maximum actual power supported for this PICPWR.

Property	Type	L	R	Description
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Sidebands	array of Connectors_PICPWR.Sidebands	Y	Y	The sidebands used by the PICPWR.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_PICPWR.Sidebands

Description: The sidebands used by the PICPWR.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this sideband.

Connectors_PhysicalUSB

Description: Information about a physical USB connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Type	string	Y	Y	The USB type of the connector for this physical USB port. Possible values: `A` (USB type A connector as described in USB `A' Plug Form Factor),` C `(USB type C connector as described in Universal Serial Bus Type-C Cable and Connector Specification),` other` (Any other connector type)
Destination	string			The endpoint location of this physical USB port. Possible values: `Internal` (A USB port inside the system chassis.), `External` (A USB port external to the system chassis that can be accessed without opening the chassis.)
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.

Property	Type	L	R	Description
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_PowerDistributionBoardManagement

Description: Information about a power distribution board management connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
ManagementType	string			The type of power distribution management board connector, such as type 1 or type 2.
Type	string			The connector type for this power distribution management board connector.
Direction	string			The intended linkage for this power distribution management board connector. Possible values: `Upstream` (Up, toward the HPM), `Downstream` (Down, toward another PDB)
Sidebands	SidebandList			A set of sidebands.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the

Property	Type	L	R	Description
				buses section of this document.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_PowerSupply

Description: Information about a power supply connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Type	string			The type of connector for this power supply.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
HotPlugSupported	boolean			Indicates whether the power supply supports hot plugging.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.

Property	Type	L	R	Description
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_RealTimeClockBattery

Description: Information about a real-time clock battery connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
PhysicalLocation	PhysicalLocation			The physical location of a component board.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.

Connectors_SCI

Description: Information about an SCI connector.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Revision	string			The revision of this SCI.
Version	string			The version of this SCI.
CommonCircuitType	string			The number of nodes supported by this SCI. Possible values: `Single Node` (One SCI node), `Dual Node` (Two SCI nodes), `Other` (Any other number of SCI nodes)
BackupBatteryVoltageMeasurementCapable	boolean			Indicates whether this SCI supports measurement of the backup battery voltage.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the buses section of this document.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
Signals	array of Signal			Information about the signals for this SCI.

Property	Type	L	R	Description
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Connectors_SOCs

Description: Information about an SOC connector.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SupportedVendorName	string			The name of the vendor that this SOC connector supports.
SupportedVendorID	string			The PCIe VendorID for the vendor that this SOC connector supports.
Type	string	Y	Y	The type of the SOC.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
DimmChannelsSupported	integer			The number of dimm channels supported by the SOC.
PhysicalLocation	PhysicalLocation			The physical location of a component board.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.

Property	Type	L	R	Description
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.

Devices

Description: Information about active and passive hardware devices

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Type	string	Y	Y	The type of the device.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
Active	boolean			Indicates whether this is an active programmable device.
Primary	boolean			Indicates whether this is the primary active programmable device.
DataFile	integer			The data file with JSON information specific to this device.
GPIOs	GPIOs	Y		Information about a set of GPIOs.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.

Property	Type	L	R	Description
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
CapacityBytes	number			The storage capacity for this device.
PhysicalSignals	array of Signal	Y	Y	Information about the physical signals for this device.
ConnectedBuses	ReferencedBusList	Y	Y	A collection of references to hardware or logical buses as defined in the buses section of this document.
OEM	Devices.OEM			Provides additional OEM-specific information for a device.

Devices.OEM

Description: Provides additional OEM-specific information for a device.

No properties defined.

Buses

Description: Information about the logical and physical buses.

Property	Type	L	R	Description
DisplayPort	array of Buses_DisplayPort			Information about DisplayPort bus.s
ESPI	array of Buses_ESPI	Y		Information about ESPI bus.s
I2C	array of Buses_I2C	Y	Y	Information about I2C bus.s
I3C	array of Buses_I3C	Y		Information about I3C bus.s

Property	Type	L	R	Description
Intrusion	array of Buses_Intrusion			Information about intrusion bus.s
JTAG	array of Buses_JTAG	Y		Information about JTAG bus.s
LTPI	array of Buses_LTPI	Y		Information about LTPI bus.s
MPESTI	array of Buses_MPESTI	Y		Information about MPESTI bus.s
NCSI_RBT	array of Buses_NCSI_RBT	Y		Information about NC-SI RBT bus.s
PCIe	array of Buses_PCIe	Y	Y	Information about PCIe bus.s
PECI	array of Buses_PECI			Information about PECI bus.s
QSPI	array of Buses_QSPI			Information about QSPI bus.s
SGMII	array of Buses_SGMII			Information about SGMII bus.s
SPI	array of Buses_SPI	Y		Information about SPI bus.s
SGPIO	array of Buses_SGPIO	Y		Information about SGPIO bus.s
UART	array of Buses_UART			Information about UART bus.s
USB	array of Buses_USB			Information about USB bus.s
OEM	array of Buses.OEM			Provides information for an OEM-specific bus.

Buses.OEM

Description: Provides information for an OEM-specific bus.

No properties defined.

Buses_DisplayPort

Description: Information about a DisplayPort bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_DisplayPort.Segments			Information about DisplayPort segments.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_DisplayPort.Segments

Description: Information about DisplayPort segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.

Property	Type	L	R	Description
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_ESPI

Description: Information about an ESPI bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
MaxClockSpeedMHz	integer			The maximum clock speed for this ESPI bus.
AlertUsage	string			The manner in which alerts are signaled on the bus. Possible values: `Embedded` (Alerts are encoded into data pins, signaling alerts via I/O[1].), `Discrete` (Alerts are encoded into a discrete signal on a dedicated alert pin.)
Density	string			The data transfer mode supported by this ESPI bus. Possible values: `Quad` (Four bi-directional I/O pins used for

Property	Type	L	R	Description
				commands and data), `Dual` (Bi-directional I/O pins used for commands and data), `Single` (Unidirectional I/O pins used for commands and data)
Controller	string			The component that receives interrupt requests and controls this bus.
Segments	array of Buses_ESPI.Segments			Information about ESPI segments.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_ESPI.Segments

Description: Information about ESPI segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_I2C

Description: Information about an I2C bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string	Y	Y	The name used in the netlist for this bus.
VoltageTenthsVolts	integer			The voltage needed to drive this bus.
RequiredPowerState	RequiredPowerState	Y	Y	The minimum power state to utilize a bus.
FrequencyKHz	integer	Y	Y	The clock frequency for this bus.
Description	string			A description of the intended usage for this bus.
Segments	array of Buses_I2C.Segments	Y	Y	Information about I2C segments.

Buses_I2C.Segments

Description: Information about I2C segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.

Property	Type	L	R	Description
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.
MUXes	array of MUX			Information about MUX.s
Bridges	array of Buses_I2C.Segments.Bridges			Information about I2C bridges in this I2C segment.

Buses_I2C.Segments.Bridges

Description: Information about I2C bridges in this I2C segment.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
Version	string			The version for this bridge.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
LinkedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the

Property	Type	L	R	Description
				buses section of this document.

Buses_I3C

Description: Information about an I3C bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
ProtocolType	string		Y	The protocol type for this I3C bus. Possible values: `Basic` (Basic (royalty-free) I3C as defined in Specification for I3C Basic, Improved Inter Integrated Circuit -- Basic), `Standard` (I3C standard as defined in Specification for I3C Basic, Improved Inter Integrated Circuit -- Basic)
VoltageTenthsVolts	integer			The voltage needed to drive this I3C bus.
RequiredPowerState	RequiredPowerState		Y	The minimum power state to utilize a bus.
FrequencyKHz	integer		Y	The frequency at which this I3C bus shall operate.
Segments	array of Buses_I3C.Segments	Y	Y	Information about I3C segments.

Buses_I3C.Segments

Description: Information about I3C segments.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
Hubs	array of Buses_I3C.Segments.Hubs	Y	Y	Information about I3C hubs.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList	Y	Y	A collection of references to connectors attached to a bus.

Buses_I3C.Segments.Hubs

Description: Information about I3C hubs.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Control	ChannelControl			Information about mechanisms used to control a MUX or hub.
Ports	array of Buses_I3C.Segments.Hubs.Ports	Y	Y	Information about I3C ports.

Property	Type	L	R	Description
PropagationDelayNS	integer		Y	The propagation delay for an I3C Hub.
Address	string	Y	Y	The I3C address at which this hub responds.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Description	string			A description of the intended usage for this hub.

Buses_I3C.Segments.Hubs.Ports

Description: Information about I3C ports.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
ChannelConfig	ChannelConfig			Information about selecting a channel in a MUX or a hub.
VoltageTenthsVolts	integer			The voltage needed to drive this I3C port.
TargetPortMode	string			The target mode for this I3C port. Possible values: `Basic` (Basic port

Property	Type	L	R	Description
				mode, as defined in the I3C specification), `Standard` (MIPI port mode, as defined in the I3C specification)
Endpoint	Endpoint			An endpoint attached to this component.
Description	string			A description of the intended usage for this port.

Buses_Intrusion

Description: Information about an intrusion bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_Intrusion.Segments			Information about intrusion segments.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_Intrusion.Segments

Description: Information about intrusion segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_JTAG

Description: Information about a JTAG bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_JTAG.Segments	Y	Y	Information about JTAG segments.

Buses_JTAG.Segments

Description: Information about JTAG segments.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList	Y		Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.
Terminated	boolean			Indicates whether this JTAG bus is terminated.

Buses_LTPI

Description: Information about an LTPI bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
RequiredPowerState	RequiredPowerState		Y	The minimum power state to utilize a bus.
MaxClockSpeedMHz	integer			The maximum clock speed for this LTPI bus.

Property	Type	L	R	Description
Segments	array of Buses_LTPI.Segments			Information about LTPI bus segments.
ConnectedDevices	ConnectedDeviceList	Y	Y	Information about devices connected to a bus.
Connectors	AttachedConnectorList	Y	Y	A collection of references to connectors attached to a bus.

Buses_LTPI.Segments

Description: Information about LTPI bus segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_MPESTI

Description: Information about an MPESTI bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_MPESTI.Segments			Information about MPESTI bus segments.
ConnectedDevices	ConnectedDeviceList	Y	Y	Information about devices connected to a bus.
Connectors	AttachedConnectorList	Y	Y	A collection of references to connectors attached to a bus.

Buses_MPESTI.Segments

Description: Information about MPESTI bus segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_NCSI_RBT

Description: Information about an NC-SI RBT bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_NCSI_RBT.Segments			Information about segments for this NC-SI RBT bus.

Buses_NCSI_RBT.Segments

Description: Information about segments for this NC-SI RBT bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
Isolation	Buses_NCSI_RBT.Segments.Isolation			Information about isolation management for this NC-SI RBT bus.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.

Property	Type	L	R	Description
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_NCSI_RBT.Segments.Isolation

Description: Information about isolation management for this NC-SI RBT bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Channels	array of Buses_NCSI_RBT.Segments.Isolation.Channels			Information about channels for isolation management.
IsolationControl	array of Buses_NCSI_RBT.Segments.Isolation.IsolationControl			Information about control for isolation management.

Buses_NCSI_RBT.Segments.Isolation.Channels

Description: Information about channels for isolation management.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Endpoint	string			The endpoint to which this channel is connnected.

Buses_NCSI_RBT.Segments.Isolation.IsolationControl

Description: Information about control for isolation management.

Property	Type	L	R	Description
Signal	string			The signal that controls this isolation mechanism.
Polarity	string			The polarity of the signal that controls this isolation mechanism. Possible values: `active_high` (Set the signal to high to disable isolation and connect to the endpoint.), `active_low` (Set the signal to low to disable isolation and connect to the endpoint.)

Buses_PCIe

Description: Information about a PCIe bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Lanes	integer	Y	Y	The number of PCIe lanes for this bus.
Version	number	Y	Y	The highest generation version of PCIe supported by this bus.
Root	string	Y	Y	The location of the PCIe root. Possible values: `Host` (The bus is rooted at the host SOC.), `BMC` (The bus is rooted at the BMC.), `Other` (The bus is not

Property	Type	L	R	Description
				rooted at either the host SOC or at the BMC.)
Fabric	Buses_PCIe.Fabric	Y	Y	Information about PCIe bus fabrics.

Buses_PCIe.Fabric

Description: Information about PCIe bus fabrics.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this fabric.
HostSOCRootPort	string			The assigned port for the SOC PCIe root complex enumeration.
FirstLane	integer			The first PCIe lane used for this fabric.
Width	integer			The number of PCIe lanes used for this fabric.
Segments	array of Buses_PCIe.Fabric.Segments			Information about segments of a PCIe bus.

Buses_PCIe.Fabric.Segments

Description: Information about segments of a PCIe bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
Switch	Buses_PCIe.Fabric.Segments.Switch			Information about the switch for a PCIe segment.
Bridges	array of Buses_PCIe.Fabric.Segments.Bridges			Information about the bridges attached to a PCIe segment.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_PCIe.Fabric.Segments.Switch

Description: Information about the switch for a PCIe segment.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
VendorID	string			The PCIe vendor ID for this PCIe switch.
SubvendorID	string			The PCIe subvendor ID for this PCIe switch.
DeviceID	string			The PCIe device ID for this PCIe switch.

Property	Type	L	R	Description
SubdeviceID	string			The PCIe subdevice ID for this PCIe switch.
RootPort	string			The root port associated with this PCIe switch.
LaneMapping	array of integer			The mapping of lanes for this PCIe switch.
HotPlugSupported	boolean			Indicates whether the PCIe switch is hot-pluggable.
DMASupported	boolean			Indicates whether the PCIe switch supports direct memory access.
CxlSupported	boolean			Indicates whether the PCIe switch supports Compute Express Link (CXL).
CxlVersionMinimum	string			The minimum version of the CXL specification supported by this PCIe switch.
LinkSpeedSupportedMaximumGBPerSec	integer			The maximum link speed supported by this PCIe switch for a single lane.
BifurcationCapable	boolean			Indicates whether the PCIe switch supports bifurcation of its PCIe lanes.
BifurcationMinimum	integer			The minimum lane width supported by this PCIe switch for bifurcation.
LatencyNS	integer			The latency of this PCIe switch.
Ports	array of Buses_PCIe.Fabric.Segments.Switch.Ports			The ports attached to this PCIe switch.
Models	Models			The possible models for this component.

Property	Type	L	R	Description
Manufacturers	Manufacturers			The possible manufacturers for this component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Buses_PCIe.Fabric.Segments.Switch.Ports

Description: The ports attached to this PCIe switch.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Width	integer			The number of PCIe lanes used for this port.
Endpoints	array of Endpoint			The endpoints attached to this port.
Type	string			The type of this port. Possible values: `Ingress` (This port is an ingress port.), `Egress` (This port is an egress port.)

Buses_PCIe.Fabric.Segments.Bridges

Description: Information about the bridges attached to a PCIe segment.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
VendorID	string			The PCIe vendor ID for this bridge.
SubvendorID	string			The PCIe subvendor ID for this bridge.
DeviceID	string			The PCIe device ID for this bridge.
SubdeviceID	string			The PCIe subdevice ID for this bridge.
RootPort	string			The root port associated with this PCIe bridge.
Width	string			The width, in PCIe lanes, for this PCIe bridge.
LaneMapping	array of integer			The mapping of lanes for this PCIe bridge.
HotPlugSupported	boolean			Indicates whether the PCIe bridge is hot-pluggable.
DMASupported	boolean			Indicates whether the PCIe bridge supports direct memory access.
CxlSupported	boolean			Indicates whether the PCIe bridge supports Compute Express Link (CXL).
CxlVersionMinimum	string			The minimum version of the CXL specification supported by this PCIe bridge.
LinkSpeedSupportedMaximumGBPerSec	integer			The maximum link speed supported by this PCIe bridge.
BifurcationCapable	boolean			Indicates whether the PCIe bridge supports bifurcation of its PCIe lanes.

Property	Type	L	R	Description
BifurcationMinimum	integer			The minimum lane width supported by this PCIe bridge for bifurcation.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
PhysicalLocation	PhysicalLocation			The physical location of a component board.
Endpoint	Endpoint			An endpoint attached to this component.

Buses_PECI

Description: Information about a PECI bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_PECI.Segments			Information about PECI bus segments.

Property	Type	L	R	Description
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_PECI.Segments

Description: Information about PECI bus segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_QSPI

Description: Information about a QSPI bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.

Property	Type	L	R	Description
NetlistName	string			The name used in the netlist for this bus.
MaxClockSpeedMHz	integer			The maximum clock speed for this QSPI bus.
Density	string			The data transfer mode supported by this ESPI bus. Possible values: `Quad` (Four bi-directional I/O pins used for commands and data), `Dual` (Bi-directional I/O pins used for commands and data), `Single` (Unidirectional I/O pins used for commands and data)
Controller	string			The component that receives interrupt requests and controls this bus.
Segments	array of Buses_QSPI.Segments			Information about QSPI segments.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_QSPI.Segments

Description: Information about QSPI segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.

Property	Type	L	R	Description
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_SGMII

Description: Information about an SGMII bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_SGMII.Segments			Information about SGMII bus segments.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_SGMII.Segments

Description: Information about SGMII bus segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_SGPIO

Description: Information about an SGPIO bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
Segments	array of Buses_SGPIO.Segments			Information about SGPIO bus segments.
ConnectedDevices	ConnectedDeviceList	Y		Information about devices connected to a bus.
Connectors	AttachedConnectorList	Y		A collection of references to connectors attached to a bus.

Buses_SGPIO.Segments

Description: Information about SGPIO bus segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_SPI

Description: Information about a SPI bus.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
MaxClockSpeedMHz	integer			The maximum clock speed for this SPI bus.
AlertUsage	string			The manner in which alerts are signaled on the bus. Possible values: `Embedded` (Alerts are encoded into

Property	Type	L	R	Description
				data pins, signaling alerts via I/O[1].), `Discrete` (Alerts are encoded into a discrete signal on a dedicated alert pin.)
Density	string			The data transfer mode supported by this ESPI bus. Possible values: `Quad` (Four bi-directional I/O pins used for commands and data), `Dual` (Bi-directional I/O pins used for commands and data), `Single` (Unidirectional I/O pins used for commands and data)
InterruptRequest	string			Indicates the component to which the IRQ[0] pin is connected.
Controller	string			The component that receives interrupt requests and controls this bus.
Segments	array of Buses_SPI.Segments			Information about SPI segments.
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_SPI.Segments

Description: Information about SPI segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_UART

Description: Information about a UART bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
VoltageTenthsVolts	integer		Y	The voltage needed to drive this UART bus.
RequiredPowerState	RequiredPowerState		Y	The minimum power state to utilize a bus.
Segments	array of Buses_UART.Segments		Y	Information about UART segments.

Buses_UART.Segments

Description: Information about UART segments.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
MUXes	array of MUX			Information about MUX.s
ConnectedDevices	ConnectedDeviceList			Information about devices connected to a bus.
Connectors	AttachedConnectorList			A collection of references to connectors attached to a bus.

Buses_USB

Description: Information about a USB bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this bus.
SupportedVersion	string		Y	The maximum supported USB version for this USB bus.
HostController	string		Y	The entity that controls the bus. Possible values: `Host` (This bus is controlled by the host SOC.), `BMC` (This bus is controlled by the BMC.)

Property	Type	L	R	Description
VoltageTenthsVolts	integer		Y	The voltage needed to drive this USB bus.
RequiredPowerState	RequiredPowerState		Y	The minimum power state to utilize a bus.
Description	string			A description of the intended usage for this bus.
Segments	array of Buses_USB.Segments			Information about segments for this USB bus.

Buses_USB.Segments

Description: Information about segments for this USB bus.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this segment.
ConnectedDevices	ConnectedDeviceList	Y	Y	Information about devices connected to a bus.
Connectors	AttachedConnectorList	Y	Y	A collection of references to connectors attached to a bus.
MUXes	array of MUX			Information about MUX.s
Hubs	array of Buses_USB.Segments.Hubs	Y	Y	Information about hubs for this USB bus segment.

Property	Type	L	R	Description
Bridges	array of Buses_USB.Segments.Bridges	Y	Y	Information about bridges for this USB bus segment.

Buses_USB.Segments.Hubs

Description: Information about hubs for this USB bus segment.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Control	ChannelControl			Information about mechanisms used to control a MUX or hub.
Ports	array of Buses_USB.Segments.Hubs.Ports			Information about ports for this USB bus hub.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Description	string			A description of the intended usage for this hub.
PhysicalLocation	PhysicalLocation			The physical location of a component board.

Buses_USB.Segments.Hubs.Ports

Description: Information about ports for this USB bus hub.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
ChannelConfig	ChannelConfig			Information about selecting a channel in a MUX or a hub.
VoltageTenthsVolts	integer			The voltage needed to drive this USB port.
Endpoint	Endpoint			An endpoint attached to this component.

Buses_USB.Segments.Bridges

Description: Information about bridges for this USB bus segment.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
Version	string			The version of USB supported by this USB bridge.
LinkedBuses	ReferencedBusList			A collection of references to hardware or logical buses as defined in the

Property	Type	L	R	Description
				buses section of this document.

Common Definitions

AttachedConnectorList

Description: A collection of references to connectors attached to a bus.

Array Item Properties:

Property	Type	L	R	Description
Type	string	Y	Y	The type of connector for this reference.
ChipSelect	number			The chip selection pin needed to access this device. This is primarily intended for use with ESPI, QSPI and SPI bus connections.
Endpoint	string	Y	Y	The endpoint connector for this reference.
MMBICapable	boolean			Whether the connector supports MMBI via the attached bus.
Width	integer			The number of lanes used for this connector.

ChannelConfig

Description: Information about selecting a channel in a MUX or a hub.

Array Item Properties:

Property	Type	L	R	Description
ControlPin	string			A pin needed to select this MUX or hub channel.

Property	Type	L	R	Description
Setting	number			The setting needed on a pin to select this MUX or hub channel.

ChannelControl

Description: Information about mechanisms used to control a MUX or hub.

Array Item Properties:

Property	Type	L	R	Description
ControlPin	string	Y	Y	A pin used to control this MUX channel.
Signal	ChannelControl.Signal	Y	Y	The signal used to control the pin.

ChannelControl.Signal

Description: The signal used to control the pin.

Property	Type	L	R	Description
Type	string	Y	Y	The type locating the signal used to control the pin.
TypeID	string	Y	Y	The specific device housing the signal used to control the pin.
Subtype	string	Y	Y	The subtype locating the signal used to control the pina pin.
SubtypeID	string	Y	Y	The specific signal used to control the pin.

ComponentAlias

Description: A secondary reference name for a component.

Type: string

ConnectedDeviceList

Description: Information about devices connected to a bus.

Array Item Properties:

Property	Type	L	R	Description
Endpoint	string	Y	Y	This instance of a connected device.
Alias	ComponentAlias			A secondary reference name for a component.
Address	string	Y	Y	The bus-specific hardware address needed to access a connected device, such as an I2C address.
SecondaryAddress	string			A secondary bus-specific hardware address needed to access a connected device, such as an I2C address.
RandomAccessCapable	boolean			Indicates whether the BMC can access memory on this device in random fashion through this connection.
ChipSelect	number			The chip selection pin needed to access this device. This is primarily intended for use with ESPI, QSPI and SPI bus connections.
HostAccessible	boolean			Indicates whether the host can access this device.

Direction

Description: The direction of a signal, virtual wire, or GPIO.

Type: string

Allowed Values: "input", "output", "bidir"

Endpoint

Description: An endpoint attached to this component.

Type: string

FlexIO

Description: A FlexIO

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this FLEXIO.

GPIOs

Description: Information about a set of GPIOs.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
StartBit	integer			The start bit for this GPIO.
WidthBits	integer			The width in bits for this GPIO.
Description	string			Describes the intended use for this GPIO.
Direction	Direction			The direction of a signal, virtual wire, or GPIO.

MUX

Description: Information about a MUX.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Subtype	string			The type of this MUX. Possible values: `BUS_MUX` (A MUX that switches bus segments.), `SIGNAL_MUX` (A MUX that switches generic named signal pairs.), `BUFFER` (A buffer/repeater device.), `TRANSCEIVER` (A transceiver device.), `LEVEL_TRANSLATOR` (A voltage level translator device.), `REDRIVER_CROSSPOINT_SWITCH` (A redriver/crosspoint switch device capable of routing multiple signal types (e.g. DisplayPort/USB3) to shared physical ports.)
SilkscreenName	SilkscreenName			The name silkscreened onto the HPM board for the component.
ReferenceDesignator	ReferenceDesignator			The reference designator used to logically describe the component in the BOM list.
Description	string			A description of this MUX.
Channels	array of MUX.Channels	Y	Y	Information about channels in this MUX.
Control	ChannelControl	Y		Information about mechanisms used to control a MUX or hub.

Property	Type	L	R	Description
InputSelector	ChannelControl			Information about mechanisms used to control a MUX or hub.
Enable	ChannelControl			Information about mechanisms used to control a MUX or hub.
Models	Models			The possible models for this component.
Manufacturers	Manufacturers			The possible manufacturers for this component.
PropagationDelayNS	integer			The propagation delay for this MUX.
PhysicalLocation	PhysicalLocation			The physical location of a component board.
IsolationCapable	boolean			Indicates whether the MUX can isolate voltage for channels.
LevelShiftingCapable	boolean			Indicates whether the MUX can shift between voltage levels.
FaultDetectionCapable	boolean			Indicates whether the MUX can detect faults.
FaultClearSignalCapable	boolean			Indicates whether the MUX can utilize a fault clear signal.
RiseTimeNS	integer			The rise time for this MUX.
FallTimeNS	integer			The fall time for this MUX.
I2CI3CAddress	string			The address for this MUX.
I2CI3CSecondaryAddress	string			The secondary address for this MUX.

MUX.Channels

Description: Information about channels in this MUX.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
ChannelConfig	ChannelConfig	Y	Y	Information about selecting a channel in a MUX or a hub.
Endpoint	Endpoint	Y	Y	An endpoint attached to this component.
VoltageTenthsVolts	integer			The voltage needed to drive this MUX channel.
IOType	string			The type of I/O used for this MUX channel. Possible values: `push-pull` (Push-pull dual-level output), `open drain` (Open drain I/O that drives to ground), `open collector` (Open collector I/O that drives high)

Manufacturers

Description: The possible manufacturers for this component.

Array of simple items.

Models

Description: The possible models for this component.

Array of simple items.

NamedComponent

Description: A component that is named so as to be referenceable.

Type: string

PhysicalLocation

Description: The physical location of a component board.

Property	Type	L	R	Description
UpperXMils	integer			The upper x-axis corner of the bounding box surrounding the component, in millimeters.
UpperYMils	integer			The upper y-axis corner of the bounding box surrounding the component, in millimeters.
LowerXMils	integer			The lower x-axis corner of the bounding box surrounding the component, in millimeters.
LowerYMils	integer			The lower y-axis corner of the bounding box surrounding the component, in millimeters.

ReferenceDesignator

Description: The reference designator used to logically describe the component in the BOM list.

Type: string

ReferencedBusList

Description: A collection of references to hardware or logical buses as defined in the buses section of this document.

Array Item Properties:

Property	Type	L	R	Description
Type	string	Y	Y	The type of bus for this reference.
Identifier	string	Y	Y	The instance of the bus type for this reference.

Property	Type	L	R	Description
SpecBusName	string			The name used in the spec to reference this bus.
Signals	array of Signal			Information about the signals for this referenced bus.

RequiredPowerState

Description: The minimum power state to utilize a bus.

Type: string

Allowed Values: "G3", "Pre-STBY", "STBY", "S5", "S0", "Other"

SidebandList

Description: A set of sidebands.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
NetlistName	string			The name used in the netlist for this sideband.

Signal

Description: Information about a signal.

Property	Type	L	R	Description
ConnectorName	string			The name of the signal as defined by a connector.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
Type	string	Y	Y	The type for this physical signal.
TypeID	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Subtype	string			The connector subtype for this physical signal.
SubtypeID	NamedComponent			A component that is named so as to be referenceable.
SubtypeIDPrefix	string			The connector subtype identifier prefix for a set of physical signals.
NetlistName	string			The Netlist name for this physical signal reference.
NetlistNamePrefix	string			The Netlist name prefix for a set of physical signals.
PhysicalPin	string			The physical pin for this physical signal reference.
PhysicalPinPrefix	string			The physical pin prefix for a set of physical signal pins.
FirstPhysicalPin	integer			The first physical pin number for a set of physical signal pins.
PhysicalPinCount	integer			The number of physical pins in a set of physical signal pins.

SilkscreenName

Description: The name silkscreened onto the HPM board for the component.

Type: string